

Problem Framing & Success Definition

AI Relationship Manager Agent for Paytm Money Limited

1. Background

Retail investing platforms are increasingly expected to provide personalized, real-time financial guidance to clients. As customer bases grow, relying exclusively on human relationship managers becomes expensive, difficult to scale, and inconsistent in response quality.

Paytm Money serves investors with diverse portfolios who frequently seek information regarding portfolio performance, stock analysis, transaction costs, market risks, and investment decisions. Many of these interactions are repetitive but require access to accurate client-specific and market-specific information.

The goal of this project is to build an AI-powered Relationship Manager (RM) Agent capable of assisting clients with investment-related queries while maintaining safety, transparency, and compliance constraints.

2. Core Problem Statement

Traditional LLM-based chatbots suffer from several limitations when deployed in financial advisory settings:

- They hallucinate portfolio information.
- They provide generic investment advice without considering client holdings.
- They lack access to live market data.
- They cannot perform financial calculations reliably.
- They may bypass safety and compliance requirements.
- They have no mechanism for transaction confirmation or auditability.

As a result, a generic chatbot cannot be trusted to operate as a financial relationship manager.

The challenge is therefore:

How can we build an AI Relationship Manager that combines the conversational capabilities of an LLM with reliable data sources, financial tools, safety guardrails, memory management, and transaction workflows?

3. Business Objectives

The system should enable clients to:

1. View and understand their investment portfolio.
2. Analyze stocks using live market data.
3. Estimate trade costs accurately.
4. Identify exchange surveillance risks (ASM/GSM).
5. Receive personalized portfolio insights.
6. Execute simulated trades only through explicit confirmation.
7. Receive transaction confirmations and audit logs.

At the same time, the system must:

- Prevent unauthorized actions.
- Refuse prohibited financial advice.
- Avoid unsupported tool usage.
- Maintain transparency regarding data sources.
- Operate reliably across multi-step interactions.

4. User Persona

Primary User

Name: Rajiv Malhotra

Age: 48

Occupation: Business Owner / Senior Executive

Investment Experience: Advanced

Risk Profile: Moderate

Assets Managed: Multi-stock equity portfolio

Typical Questions

- "Show me my portfolio performance."
- "Which stock is performing the worst?"
- "Should I buy more HDFCBANK?"
- "How much will this trade cost?"
- "Are there any exchange restrictions on this stock?"
- "What happens to my cash balance if I execute this trade?"

5. Project Scope

The AI RM Agent supports:

Portfolio Intelligence

- Portfolio holdings
- Allocation analysis
- Sector exposure

- Profit and loss tracking

Market Intelligence

- Live stock prices
- Volume trends
- 52-week high/low analysis
- Analyst report retrieval through RAG

Trade Assistance

- Trade cost estimation
- Cash availability checks
- Confirmation workflow
- Order execution through backend APIs

Safety & Compliance

- Explicit confirmation gate
- Guardrail enforcement
- F&O refusal
- Return-prediction refusal
- Unsupported tool rejection

Personalization

- Client-specific portfolio retrieval
- Conversation memory
- Context compression for long sessions

7. Success Metrics

The project is considered successful if the agent can:

Functional Success

- Correctly invoke required tools.
- Retrieve live portfolio data.
- Retrieve current market information.
- Calculate trade costs accurately.

Safety Success

- Refuse prohibited requests.
- Prevent execution without confirmation.

- Reject unsupported actions.
- Avoid guaranteed-return claims.

Reliability Success

- Complete multi-tool workflows.
- Handle errors gracefully.
- Maintain conversational context.

Evaluation Success

- Pass automated test cases across:
 - Portfolio Analysis
 - Trade Cost Calculation
 - Market Data Retrieval
 - Safety Guardrails
 - Multi-Tool Reasoning

Target evaluation score: **80%+ pass rate** across all categories.

8. Expected Impact

The AI Relationship Manager demonstrates how LLMs can be combined with retrieval systems, financial APIs, tool-calling frameworks, memory systems, and safety guardrails to create a trustworthy advisory assistant.

The resulting system moves beyond a traditional chatbot and towards a production-style financial assistant capable of supporting real investment workflows while maintaining transparency, safety, and user control.